

Modeling & Control of Boost Converter

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Abstract

Deriving the model for a boost converter, using it to simulate its behavior and derive a suitable control strategy for it.

1 Boost Converter Overview

A synchronous boost converter, shown in Figure 1, consists of a voltage supply V_{in} , inductor L with parasitic series resistance R_L , two switches in a half H-bridge configuration S_{high} and S_{low} , a capacitor C and finally the load R .

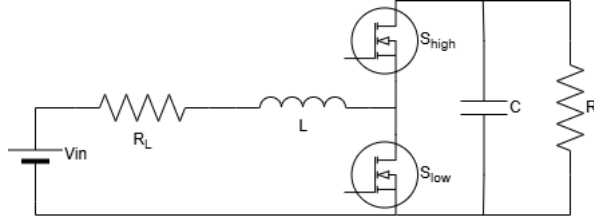


Figure 1: Boost converter schematic.

Its goal is to increase the voltage over the output load, R in this case, compared to V_{in} . This is achieved by switching S_{high} and S_{low} , resulting in the circuits shown in Figure 2 and 3.

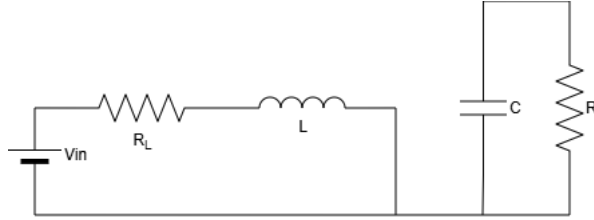


Figure 2: Boost converter schematic with S_{low} switch enabled.

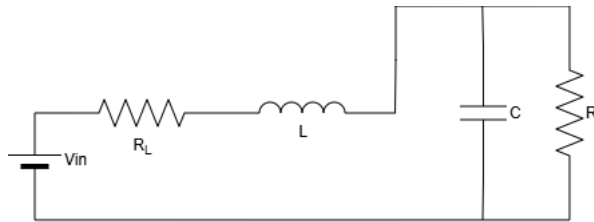


Figure 3: Boost converter schematic with S_{high} switch enabled.

TODO add Interleaved Synchronous Boost Converter, also measured voltages and currents

2 Component choices

TODO add choices for components

2.1 Battery

An 18650 battery is used, exact type to be determined. Note the requirement for high current capability.

2.2 MOSFETs

4x CSD18540Q5B for two half H-bridges. Chosen for their low $R_{ds,on}$, gate charge and price/performance.

2.3 Gate drivers

TODO

2.4 Gate resistors and diodes

TODO

2.5 Inductors

2x FC-SE2915-2R2M coils, one for each phase of the interleaved converter. This inductor supports high current capacity at a relatively low price. It has $1.86m\omega$ resistance, $2.2\mu H \pm 20\%$ inductance and supports I_{RMS} of 32A, $I_{Sat} > 100A$. Furthermore, it has a flat copper wire coil for low losses at high frequency.

2.6 Output capacitors

The output ripple of a boost converter is roughly:

$$\Delta V = \frac{I_{out} \cdot Duty}{f_{sw} \cdot C_{output}} \quad (1)$$

with $I_{out} = 3A$, $D = 0.8$ and $f_{sw} = 250kHz$. With a desired maximum output ripple of $50mV$ this results in:

$$0.05 = \frac{3A \cdot 0.8}{250kHz \cdot C_{output}}. \quad (2)$$

In conclusion, $C_{output} = 192\mu F$ or higher.

Each phase of the converter gets 5x FS55X106K500EGG MLCC $10\mu F \pm 10\%$, 50V, X7R in a 2220 package to suppress the voltage spikes from switching. At 15 – 20V, it is expected that the capacity derating is around 30 – 50%, leaving 5 – $7\mu F$ per MLCC capacitor. This results in $50\mu F$ of capacity from the

MLCC capacitors and $440\mu F$ from the polymer caps. In total around $500\mu F$ of capacity, aiming for $20mV$ of voltage ripple.

2.7 Input capacitors

Same capacitors as output are selected for first prototype, more capacity than needed. Experimentation with removing capacitors is possible.

2.8 Current sensors

To measure inductor currents, the INA240A1 is chosen, in combination with a $5m\Omega$, because of its higher bandwidth compared to HAL current sensors. The drawback is higher losses because of added resistance.

INA240A1 has a gain of $20V/V$, and an offset of $1.65V$. Using Ohm's law, $V = I \cdot R$, we can compute the range of the current sensor.

2.9 Microcontroller

STM32G431 is selected, contains USB-C PD support, various timers and other useful peripherals.

3 Deriving model

The model is derived based on the position of the switches, resulting in two models: one with the low side switch closed, and one with the high side switch closed.

Starting with the low side switch closed, the Kirchhoff's circuit laws (voltage and current) are used to describe the behavior of the network:

$$\begin{aligned} v_{in}(t) - v_{R_L}(t) - v_L(t) &= 0 \\ v_C(t) - v_R(t) &= 0 \\ i_{in}(t) &= i_{R_L}(t) = i_L(t) \\ i_C(t) + i_R(t) &= 0 \end{aligned} \tag{3}$$

Now with the high side switch closed:

$$\begin{aligned} v_{in}(t) - v_{R_L}(t) - v_L(t) - v_C(t) &= 0 \\ v_C(t) - v_R(t) &= 0 \\ i_{in}(t) &= i_{R_L}(t) = i_L(t) \\ i_L(t) - i_C(t) - i_R(t) &= 0 \end{aligned} \tag{4}$$

Next, these models are converted into a set of first order differential equations, such that it can be transformed into a state space model.

Quick reminder that:

$$\begin{aligned} v_L(t) &= L \frac{di_L(t)}{dt} \\ i_C(t) &= C \frac{dv_C(t)}{dt} \\ v_R(t) &= i_R(t)R \end{aligned} \tag{5}$$

Since there are two components that are able to store energy (capacitor and inductor), the minimal realization is of order 2. The variables of interest are the capacitor voltage $v_C(t)$ and inductor current $i_L(t)$. The two differential equations of interest are:

$$\begin{aligned} \frac{di_L(t)}{dt} &= \frac{1}{L} v_L(t) \\ \frac{dv_C(t)}{dt} &= \frac{1}{C} i_C(t) \end{aligned} \tag{6}$$

Now, for the situation with the low side switch closed, substituting

$$\begin{aligned} v_L(t) &= v_{in}(t) - v_{R_L}(t) \\ i_C(t) &= i_L(t) - i_R(t) \end{aligned} \tag{7}$$

results in

$$\begin{aligned} \frac{di_L(t)}{dt} &= \frac{1}{L} (v_{in}(t) - v_{R_L}(t)) \\ \frac{dv_C(t)}{dt} &= \frac{1}{C} (i_L(t) - i_R(t)) \end{aligned} \tag{8}$$

where $v_{R_L}(t) = R_L i_L(t)$, since $i_R(t) = i_L(t)$, and $i_R(t) = \frac{1}{R} v_C(t)$, since $v_R(t) = v_C(t)$, resulting in:

$$\begin{aligned}\frac{di_L(t)}{dt} &= \frac{1}{L}(v_{in}(t) - R_L i_L(t)) \\ \frac{dv_C(t)}{dt} &= \frac{1}{C}(i_L(t) - \frac{1}{R} v_C(t))\end{aligned}\tag{9}$$

Furthermore, the first order differential equations for when the high side switch is closed:

$$\begin{aligned}\frac{di_L(t)}{dt} &= \frac{1}{L} v_L(t) \\ \frac{dv_C(t)}{dt} &= \frac{1}{C} i_C(t)\end{aligned}\tag{10}$$

substituting

$$\begin{aligned}v_L(t) &= v_{in}(t) - v_{R_L}(t) - v_C(t) \\ i_C(t) &= i_L(t) - i_R(t)\end{aligned}\tag{11}$$

resulting in

$$\begin{aligned}\frac{di_L(t)}{dt} &= \frac{1}{L}(v_{in}(t) - v_{R_L}(t) - v_C(t)) \\ \frac{dv_C(t)}{dt} &= \frac{1}{C}(i_L(t) - i_R(t))\end{aligned}\tag{12}$$

where $v_{R_L}(t) = R_L i_L(t)$ and $i_R(t) = \frac{1}{R} v_C(t)$ resulting in:

$$\begin{aligned}\frac{di_L(t)}{dt} &= \frac{1}{L}(v_{in}(t) - R_L i_L(t) - v_C(t)) \\ \frac{dv_C(t)}{dt} &= \frac{1}{C}(i_L(t) - \frac{1}{R} v_C(t))\end{aligned}\tag{13}$$

3.1 State Space Model

Now, the equations are transformed into a state space model, which has the form:

$$\begin{aligned}\dot{x}(t) &= Ax(t) + Bu(t) \\ y(t) &= Cx(t) + Du(t)\end{aligned}\tag{14}$$

where x is the state vector with its derivative $\dot{x}$, u the input and y the output vector. Defining the relations between input, state and output are the matrices A, B, C and D . For ease of notation, the time variable (t) is omitted from now on.

The state space model when the low side switch is on:

$$\begin{aligned}\begin{bmatrix} \frac{di_L}{dt} \\ \frac{dv_C}{dt} \end{bmatrix} &= \begin{bmatrix} -\frac{R_L}{L} & 0 \\ 0 & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} [V_{in}] \\ [V_{out}] &= [0 \quad 1] \begin{bmatrix} i_L \\ v_C \end{bmatrix} + [0] [V_{in}]\end{aligned}\tag{15}$$

State space model when the low side is off, and high side is on:

$$\begin{aligned} \begin{bmatrix} \frac{di_L}{dt} \\ \frac{dv_C}{dt} \end{bmatrix} &= \begin{bmatrix} -\frac{R_L}{L} & -\frac{1}{L} \\ \frac{1}{C} & -\frac{1}{RC} \end{bmatrix} \begin{bmatrix} i_L \\ v_C \end{bmatrix} + \begin{bmatrix} \frac{1}{L} \\ 0 \end{bmatrix} [V_{in}] \\ [V_{out}] &= [0 \quad 1] \begin{bmatrix} i_L \\ v_C \end{bmatrix} + [0] [V_{in}] \end{aligned} \tag{16}$$

4 MATLAB simulation

To simulate the system the following MATLAB code is used:

```
%% State space model
% Mode 1: Switch ON (inductor charging)
A_on = [-R_L/L, 0; 0, -1/(R*C)];
B_on = [1/L; 0];

% Mode 2: Switch OFF (inductor discharging)
A_off = [-R_L/L, -1/L; 1/C, -1/(R*C)];
B_off = [1/L; 0];

%% Open loop simulation
% Simulate for N switching periods
opts = odeset('RelTol',1e-9,'AbsTol',1e-9); %
      Tolerance options
[t, x] = ode45(@(t,x) boost_dyn(t,x,A_on,B_on,A_off,
      B_off,Vin,fsw,duty), [0 t_final], x0, opts);

%% Differential equation function
function dx = boost_dyn(t, x, A_on, B_on, A_off, B_off
, Vin, fsw, duty)

    T = 1/fsw; % Switching period
    t_mod = mod(t, T); % Position inside
                     switching cycle

    if t_mod < duty*T
        % Switch ON mode
        A = A_on;
        B = B_on;
    else
        % Switch OFF mode
        A = A_off;
        B = B_off;
    end

    dx = A*x + B*Vin;
end
```


5 Control strategy

Starting with the control objective.

5.1 Objective

Control the voltage output of the synchronous boost converter between 5 and 20V with various loads, while focussing on stability and robustness.

5.2 Strategies

The synchronous boost converter is a switching system.